Technical specification template: Protection Relays, Instrument Transformers and Coordination

OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A5. Generated 2026-11-05 07:50 UTC. Package version 0.1.0.dev0.

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Document control

Field	Value
Document number	OHM-DC-2000-SPEC-09
Package revision	A5
Issue	P1
Status	DRAFT TEMPLATE
Prepared by	[name]
Checked by	[name]
Approved by	[name]
Approval date	[date]

1. Scope

This template defines the technical requirements for protection relays, instrument transformers and coordination for the OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. It is to be completed and issued by the project. Items in [square brackets] are decisions or data the project must supply. Preliminary values from the Ohmetry reference configuration are shown with their data status and must be replaced or confirmed.

2. Applicable standards

- IEC 60255 series (measuring relays and protection equipment)
- IEC 61869 series (instrument transformers)
- IEC 61850 series (communication), if adopted
- IEC 60947-2 (LV circuit-breakers)
- Editions and applicability of every standard above are NOT VERIFIED. The project shall confirm the current editions and any national deviations.

3. Service conditions and system data

Item	Requirement or preliminary value	Data status
Nominal LV system	400 V line-to-line, 60 Hz, 3-phase	PRELIMINARY_ASSUMPTION. Utility confirmation is the owner's statement, not independently verified
MV supply	13.8 kV	PRELIMINARY_ASSUMPTION. To be confirmed with the utility
Design ambient temperature	30 degC	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Site altitude	0 m	PRELIMINARY_ASSUMPTION. Site data to be confirmed

Prospective fault level	[To be calculated: short-circuit study pending utility, transformer and generator data]	NOT CALCULATED

4. Equipment ratings

Ratings below are preliminary values from the master configuration unless marked otherwise. They are not manufacturer-confirmed and are not an equipment selection.

Item	Requirement or preliminary value	Data status and note
Protective devices in scope	MV-CB-A/B, ACB-TX-A/B, ACB-GEN-A/B, ACB-UPS-IN-A/B (illustrative list)	Illustrative. Confirm with the design
Required breaking capacity	[At least the fault level at each location]	NOT CALCULATED
CT and VT ratios, classes, burdens	[To be determined by the project]	Pending
Settings	[From the approved coordination study]	NOT STARTED: needs device curves and a validated network study

5. Technical requirements

5.1 A protection coordination study shall be performed by the project with the manufacturers' device curves and a validated network model; the study, the settings schedule and the time-current characteristics shall be issued for review.

5.2 The study shall demonstrate selectivity where the project requires it and report where it is not achieved.

5.3 Minimum fault current sensitivity, earth-fault protection and the effect of generator fault contribution shall be assessed.

5.4 Arc-flash assessment: [project to define].

5.5 Relay functions, communication, time synchronisation and cyber security: [project to define].

6. Manufacturer data required

The tenderer shall supply the datasheets and values below with its offer, each with the document name and revision. A value without a source document is not accepted. These are also tracked in the manufacturer-data register and are imported into the calculations through the data override mechanism.

Parameter id	Parameter	Datasheet	Current status
distribution.MV-CB-A.breaking_capacity_ka	MV-CB-A: breaking_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.MV-CB-A.making_capacity_ka	MV-CB-A: making_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.MV-CB-A.time_current_curve_r	MV-CB-A: time_current_curve_ref	Circuit breaker or relay data sheet with time-	MISSING

ef		current curves and test certificates	
distribution.MV-CB-A.settings	MV-CB-A: settings	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-TX-A.breaking_capacity_ka	ACB-TX-A: breaking_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-TX-A.making_capacity_ka	ACB-TX-A: making_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-TX-A.time_current_curve_ref	ACB-TX-A: time_current_curve_ref	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-TX-A.settings	ACB-TX-A: settings	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-GEN-A.breaking_capacity_ka	ACB-GEN-A: breaking_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-GEN-A.making_capacity_ka	ACB-GEN-A: making_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-GEN-A.time_current_curve_ref	ACB-GEN-A: time_current_curve_ref	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-GEN-A.settings	ACB-GEN-A: settings	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-UPS-IN-A.breaking_capacity_ka	ACB-UPS-IN-A: breaking_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-UPS-IN-A.making_capacity_ka	ACB-UPS-IN-A: making_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-UPS-IN-A.time_current_curve_ref	ACB-UPS-IN-A: time_current_curve_ref	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-UPS-IN-A.settings	ACB-UPS-IN-A: settings	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING

7. Inspection and quality requirements

7.1 Review of relay datasheets, firmware versions and the settings schedule.

7.2 Inspection of panel wiring and test blocks.

7.3 Routine tests on every unit as required by the applicable standard and this specification.

7.4 Type-test reports or evidence for the design, issued by an accredited or recognised laboratory [project to define acceptable evidence].

7.5 Inspection and test plan (ITP) approved before manufacture, naming witness and hold points.

7.6 Calibration certificates for test instruments.

8. Factory acceptance test (FAT) requirements

8.1 Relay functional and secondary injection tests against the settings schedule.

8.2 Scheme and interlock tests.

8.3 Witnessing: [owner / independent inspector to attend].

8.4 A FAT procedure shall be submitted for approval before the test; the results shall be issued as a signed report, with deviations and their disposition recorded.

9. Commissioning requirements

9.1 Secondary injection of every protection function; primary injection where specified.

9.2 Trip tests through the circuit-breakers; settings verified against the approved study and recorded.

9.3 Coordination evidence retained with the commissioning records.

9.4 Commissioning records shall name the test, the instrument and its calibration, the date and the person, and shall be retained with the verification register.

10. Submittals and documentation

- Datasheets, drawings and type-test reports.
- Inspection and test plan, FAT procedure and report.
- Installation, operation and maintenance manuals.
- Spare parts and special tools lists.
- Warranty terms [project to define].

11. Open items (NOT VERIFIED)

- All ratings in section 4 marked PRELIMINARY or UNVERIFIED are open until confirmed from manufacturer data and the project studies.
- The prospective fault level is NOT CALCULATED until the utility, transformer and generator data are supplied.
- Standards editions and local requirements are NOT VERIFIED.
- This template has not been reviewed by an independent engineer.